

David Mis

PhD Candidate

Department of Computational Applied Mathematics and Operations Research

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I study mathematical machine learning for science and inverse problems, with a particular interest in measure-theoretic approaches.

Education

Rice University PhD, <i>Computational and Applied Mathematics</i> Houston, TX, USA Advised by Dr. Maarten de Hoop	2021–present
University of Texas MS, <i>Computer Science</i> Austin, TX, USA	2019–2021
University of Texas BS, <i>Computer Science</i> BS, <i>Mathematics</i> BA, <i>Economics and Chinese Language</i> Austin, TX, USA	2009–2014

Publications

Preprints

Function graph transformers universally approximate operators between function spaces T Furuya, <u>SD Mis</u> , I Dokmanić, MV de Hoop, M Lassas <i>Preprint, arXiv:2605.17968</i> arXiv · PDF	May 2026
Forward and inverse problems for measure flows in Bayes Hilbert spaces <u>SD Mis</u> , MV de Hoop <i>Preprint, arXiv:2603.20329</i> arXiv · PDF	Mar 2026
Hybrid operator learning of wave scattering maps in high-contrast media A Balaji, T Teolis, <u>SD Mis</u> , JA Lara Benitez, C Wang, MV de Hoop <i>Preprint, arXiv:2602.11197</i> arXiv · PDF	Jan 2026

Conference papers

Semialgebraic Neural Networks: From roots to representations May 2025
SD Mis, M Lassas, MV de Hoop
International Conference on Learning Representations (ICLR) 2025
Singapore
[arXiv](#) · [PDF](#) · [Conference](#)

Journal papers

Reduced mRNA secondary-structure stability near the start codon indicates functional genes in prokaryotes 2012
TE Keller, SD Mis, KE Jia, CO Wilke
Genome Biology and Evolution, Volume 4, Issue 2, Pages 80–88
[Journal](#) · [PDF](#)

Presentations

Forward and inverse problems for measure flows in Bayes Hilbert spaces Mar 2026
SIAM UQ, Minneapolis, Minnesota

Measure flows in Bayes Hilbert spaces Dec 2025
Inverse Days, Helsinki, Finland

Measure flows in Bayes Hilbert spaces: Particle methods and sampling (poster) Dec 2025
CBMS Conference, University of Houston

Diffeomorphism invariant measure flows in Bayes Hilbert spaces Jul 2025
Applied Inverse Problems (AIP) 2025, Rio de Janeiro, Brazil

Semialgebraic Neural Networks: From roots to representations (poster) May 2025
International Conference on Learning Representations (ICLR) 2025, Singapore

Semialgebraic Neural Networks: From roots to representations Dec 2024
Inverse Days, Oulu, Finland

Geo-Mathematical Imaging Group (GMIG) Partners Meetings

Function graph transformers universally approximate operators between function spaces May 2026
Rice University

Measure flows in Bayes Hilbert spaces: Diffeomorphism invariance, optimal control, and flocking Sep 2025
Rice University

Semialgebraic Neural Networks Nov 2024
Rice University

Experience

Teaching

Rice University 2021–present

Grader / Teaching Assistant

Houston, TX

Grader: Differential Equations for Science and Engineering, Partial
Differential Equations I, Applied Functional Analysis, Numerical Analysis

Graduate Teaching Assistant: Matrix Analysis

The Art of Problem Solving 2020–2021

Grader / Teaching Assistant

Virtual

Various high school mathematics courses

University of Texas 2011–2012

Undergraduate Teaching Assistant

Austin, TX

Courses: Computational Biology, Statistics for Biologists

Research

Rice University 2022–present

PhD Research Assistant

Houston, TX

Geo-Mathematical Imaging Group (GMIG)

University of Texas 2010–2012

Undergraduate Research Assistant

Austin, TX

Lab of Dr. Claus O. Wilke, Professor of Integrative Biology

Software Development

David Mis Software LLC 2018–2021

Software Developer

Houston, TX

Independent software consultant

Unaliwear 2015–2018

Cloud Engineer

Austin, TX

Principal architect of AWS infrastructure for consumer IOT device

Hewlett Packard
Software Developer
Austin, TX
Enterprise software development

2014–2015

Service and Awards

Professional Activities

Minisymposium Organizer
Measure Flows for Inverse Problems and Machine Learning
SIAM UQ '26

March 2026

Mentorship

Rice Data2Knowledge Capstone Project Mentor
Rice University
NASA Orbital Transfer Project

Spring 2025

Awards

SIAM UQ 2026 student travel award